

- 3 (a) (i) When the gas molecules that are in continuous random motion collides with the inner wall of the bubble and rebounds, there is a change in velocity and hence a change in momentum. **B1**

By Newton's Second Law, the bubble walls will exert a force on the gas molecules. By Newton's Third Law, the gas molecules will exert an equal an opposite force on the inner walls of the bubble. **B1**

The force per unit area exerted by the gas molecules on the inner walls

of the bubble gives rise to the pressure of the bubble.

B1

MC:	Generally, the question was not very well done despite being rather lenient in the marking. Many responses were essentially a regurgitation of the answers that was used in Junior High/O-level without bringing in concepts learnt at the A-levels. Students need to be aware that the level and depth of their response must evolve in proportion to the level of the exam that they are sitting for. <u>Some common misconceptions/issues include:</u> - Gas molecules moving around and colliding <u>with each other</u> which causes a change in momentum (this is a violation of ideal gas assumption and also the wrong reason for the change in momentum of gas particles resulting in pressure) - Pressure was due to the collision of the gas molecules <u>with water molecules</u> (Pressure of gas is due to collisions of gas molecules with the inner walls of its container – appropriate terminologies should be used in explanations) <u>No explicit links made to Newton's 3rd law</u> in their explanations. Many jumped straight to stating that since the gas molecules experienced a change in momentum, the gas molecules exert a force on the inner walls of the bubble. - There is a change in momentum of the inner walls of the bubble when the gas molecules collide (Note that while this is technically not a misconception, since $m_{\text{wall}} \gg m_{\text{particle}}$, by using conservation of momentum, $v_{\text{wall}} \approx 0$ both before and after collision. - Therefore, it is more meaningful to centre the discussion based on the gas molecules.)
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(ii) Since temperature constant:

$$\begin{aligned}
 p_1 V_1 &= p_2 V_2 \\
 p_1 \left(\frac{4}{3} \pi r^3 \right) &= (p_1 - h \rho g) (V_2) \\
 (2.4 \times 10^5) \left[\frac{4}{3} \pi (0.015)^2 \right] &= [2.4 \times 10^5 - 14(1000)(9.81)] V_2 & \mathbf{C1} \\
 V_2 &= 3.3050 \times 10^{-5} \text{ m}^3 \\
 &= 3.3 \times 10^{-5} \text{ m}^3 & \mathbf{A1}
 \end{aligned}$$

MC:	Poorly done. Common (more eye-catching) issues include: - Erroneously assuming $P_{\text{atm}} = 101325 \text{ Pa}$ or $1.01 \times 10^5 \text{ Pa}$. Values were given in the question to calculate the pressure of the atmosphere in this context. (Very common mistake) - Wrong formula used for the volume of sphere. A number of candidates used πr^2 instead. - Using diameter instead of radius to compute initial volume.
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- (b) (i) The First Law of Thermodynamics states that the increase in internal energy of a system **B1**
is the sum of the heat supplied to the system and the work done on the system. **B1**

MC:	Very commonly tested definition, which has very specific keywords that have been clearly outlined by the syllabus outcomes. Hence close to zero variance in answers will be accepted. (except minor things like swapping the order of Q and W in the definition) Common mistakes: • <u>Change or total</u> in internal energy • Heat supplied to <u>the gas</u> / work done on <u>the gas</u> (First law of thermodynamics applies to systems that are non-gaseous as well, for eg. Electrical circuits)
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- (ii) Since the bubble has expanded at a constant temperature, the work done on the gas is negative and the increase in internal energy of the gas is zero. **B1**

Hence, by the First Law of Thermodynamics, heat is added into the system.

B1

MC:	Was decently done. Most students who applied the first law of thermodynamics were able to arrive at the correct conclusion. Those who tried using $PV = nRT$ to were generally not successful in gaining credit.
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- (iii) The pressure of the bubble decreases. **A1**
A non-ideal gas has non-negligible intermolecular forces of attraction which reduces the average force exerted by the gas molecules on the walls of the container. **B1**

Do not Accept: Non-ideal gas will collide inelastically with the walls of the container and hence average force exerted by gas molecules to decrease.

MC:	Was decently done. Most students were able to state that the intermolecular forces of attraction would cause the pressure to decrease. Those who attained partial credit generally did not make the link between how intermolecular forces of attraction will cause pressure to decrease.
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4 (a) (i) From Fig. 4.2 amplitude decreases with distance from the source **B1**